

Table 4: Source accuracies at required target accuracy[%] of each method.

|                                                     |              |              |              |
|-----------------------------------------------------|--------------|--------------|--------------|
| Source task                                         | CIFAR-10     | CIFAR-100    | CIFAR-10     |
| Target task                                         | CIFAR-100    | CIFAR-10     | SVHN         |
| Required target accuracy (w.r.t. fine-tuning model) | 65.30 (95%)  | 88.04 (95%)  | 91.17 (95%)  |
| LwF (?)                                             | 89.59        | 63.91        | 38.08        |
| ResCL (Ours)                                        | <b>90.65</b> | <b>68.13</b> | <b>76.83</b> |

Table 5: Source accuracies at required target accuracy[%] of each method. The second column represents sequential learning on three tasks.

|                                                     |                      |              |              |
|-----------------------------------------------------|----------------------|--------------|--------------|
| Source task                                         | CIFAR-10 → CIFAR-100 | ImageNet     |              |
| Target task                                         | SVHN                 | CUB          |              |
| Network architecture                                | preResNet            | AlexNet      | VGG          |
| Required target accuracy (w.r.t. fine-tuning model) | 90.07 (95%)          | 50.83 (95%)  | 67.04 (95%)  |
| LwF (?)                                             | 41.84                | 40.17        | 66.55        |
| Mean-IMM (?)                                        | —                    | 52.33        | 68.48        |
| ResCL (Ours)                                        | <b>53.24</b>         | <b>53.91</b> | <b>69.73</b> |

tion parameters with respect to the depth of their layers. For all scenarios in Fig. 6, the magnitude of the changes tends to increase with the depth. In deep convolutional neural networks, it is known that shallow layers learn basic features such as colors, edges, and corners, whereas deep layers learn class-specific features such as dog faces and bird legs (?). Thus, shallow layers do not need to be changed much since their features are already common in both tasks, but deep layers have large deviations from the original ones since the classes of the target task are different from those of the source task.

## Conclusion

We have proposed a novel continual learning method, ResCL, which exhibits the state-of-the-art performance for continual learning of image classification tasks. It prevents catastrophic forgetting, even if the source and target tasks are very different. ResCL can be used in practice, as no information about the source task is required, except the orig-

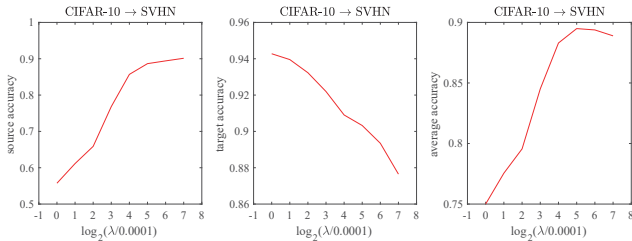


Figure 5: Source, target, and average accuracies with respect to the trade-off hyperparameter  $\lambda$  in the CIFAR-10 → SVHN scenario.

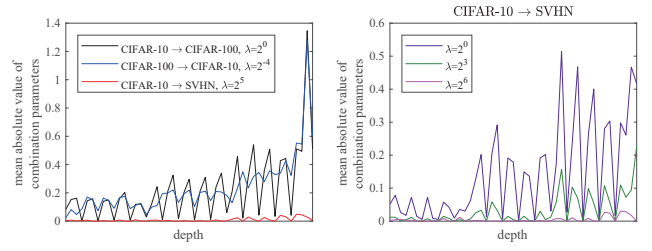


Figure 6: Mean absolute value of the elements of the combination parameter  $\alpha$  with respect to the depth.

inal network, and the size of a network does not increase in the inference phase. Moreover, any general CNN architectures can be adopted since our method is designed to handle convolution and BN layers.

In this study, we limited the scope of the task to sequential learning of image classification with CNNs. However, the ResCL method can be naturally extended to support other types of neural networks, such as recurrent neural networks, since it simply linearly combines the outputs of two layers. We leave the application of the ResCL method to other fields for future work.

## Acknowledgment

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